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A novel O2O service recommendation method based on dynamic preference similarity☆

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ABSTRACT

Recent technological advancements have enabled an increasing number of consumers to select services from online platforms and utilize them in offline stores, a model known as online-to-offline (O2O) e-commerce. This emerging model has garnered significant attention from both business and academic communities. However, with the rapid growth of O2O services, consumers face challenges in selecting services that align with their preferences from a vast array of options. To address this issue, this paper proposes a novel O2O service recommendation method based on dynamic similarity estimation (ReDPS). The dynamic similarity is calculated by tracking changes in consumer preferences over time, providing a more accurate and robust measure of consumer relationships. We validate the ReDPS method using both the Dianping dataset and the publicly available Yelp dataset. Experimental results show that: 1) ReDPS significantly outperforms classical and state-of-the-art recommendation methods, with its effectiveness improving over longer time spans of consumer feature data. 2) Consumer preferences are more strongly influenced by variations in service categories and geographical locations over time than by changes in service evaluations, though all factors are important, and consumers of the same gender tend to exhibit similar preferences. 3) Optimal parameter configurations for ReDPS are identified through the experiments.

1. Introduction

In recent years, the rise of the online-to-offline (O2O) business model has been driven by the growing popularity and widespread adoption of various O2O services, such as group-buying for dining, ride-hailing, bike-sharing, and movie ticketing [1]. Unlike traditional e-commerce platforms, such as Amazon, which aim to shift consumer shopping behavior from offline to online [2], O2O service platforms are closely integrated with brick-and-mortar businesses, acting as digital intermediaries that connect consumers with merchants through online purchases followed by offline consumption [3]. For instance, in catering group-buying—the most popular O2O service in China [4]—consumers first select and purchase a coupon through an O2O platform, such as Dianping, and then redeem it at offline restaurants to enjoy the corresponding deal.

After consumption, consumers can evaluate services on O2O platforms, similar to how they provide ratings and reviews on platforms like Amazon and Tmall. On these e-commerce platforms, consumers often struggle to find products that align with their interests and needs due to the overwhelming number of available items, leading to the issue of information overload [5]. This challenge is also prevalent on O2O service platforms. For example, Dianping, one of the leading O2O platforms in China, offers more than 200 types of services across approximately 2800 cities and counties [6]. In Beijing alone, according to data from 2016 to 2020, as many as 23,009 restaurants offered coupons on the platform. As a result, consumers face difficulties in locating their "ideal" coupons on O2O platforms, much like the challenges they encounter on traditional e-commerce platforms.

Recommender systems have proven effective in addressing the issue of information overload [7,8]. For instance, over 35 % of Amazon's sales and more than 60 % of Netflix's revenue are driven by their accurate personalized recommendation systems [9]. These systems aim to suggest items to a target consumer based on their preferences, such as interests and needs, as well as the consumption patterns of other consumers with

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similar preferences [10]. Thus, the accuracy of similarity estimation, derived from consumer preferences, plays a critical role in the recommendation process [11]. It is important to note that consumer preferences are dynamic, as their interests and needs evolve over time [12,13]. However, existing similarity estimation methods are based on static data and fail to account for temporal changes, leading to inaccuracies [14]. Additionally, tracking similar shifts in consumer preferences over time can indicate stronger relationships between consumers [15]. For instance, a consumer is more likely to select items recommended by peers who demonstrate high dynamic similarity in preferences.

Therefore, dynamic similarity not only enhances accuracy but also strengthens consumer relationship analysis, significantly improving recommendation performance. While consumer preference shifts are shaped by various complex factors such as age, income, family, and social circles, these changes are reflected through differences in service selection, evaluation, and location across both adjacent and distinct time periods [15]. In this study, we propose a novel dynamic similarity model based on consumer preference changes, captured through variations in service selection, evaluation, and location. The recommendation process begins by calculating dynamic similarities between all consumers. For each target consumer, we identify neighbors with strong relationships based on these dynamic similarities. We then prioritize and recommend O2O services by considering both the strength of these relationships and the neighbors' evaluations of services the target consumer has yet to try. The key contributions of this work are as follows:

(1) We propose a novel dynamic similarity estimation method for O2O service recommendations, based on changes in consumer preferences over time. These changes are reflected in service selections, evaluations, and locations, with service location being particularly significant in the O2O business model, distinguishing it from traditional e-commerce models.

(2) We develop new approaches to measure preference changes between two consumers over time across the three key dimensions: service selection, evaluation, and location. These aspects are then integrated to generate a dynamic similarity score, which is used to rank consumer neighbors and recommend relevant O2O services for the target consumer.

(3) In our experiments, we confirm that the proposed recommendation method, ReDPS, consistently outperforms both traditional and state-of-the-art methods in terms of recommendation performance. We also examine the optimal parameter values of the method and their managerial implications. Furthermore, we investigate the impact of consumer gender on O2O service recommendations, revealing that consumers of the same gender tend to exhibit similar preferences in specific domains.

The remainder of this paper is organized as follows: The second section offers a concise review of the literature on the O2O business model, personalized recommendation systems, and similarity estimation. The third section describes the methodology for dynamic similarity estimation based on changes in consumer preferences. The fourth section outlines the process of recommending O2O services using this innovative similarity approach. The fifth section presents the experimental results and their implications, and the paper concludes in the sixth section.

2. Literature review

2.1. O2O business model

Online to Offline (O2O) business model commands intense attention from both academic and practical fields [17]. In this business model, consumers search services online but then completing their consumption in a physical store [18,19]. Given that online-offline integration is a popular phenomenon, it is not surprising there are enough literatures that studies it. Li et al. investigate the factors that affect the growth of Groupon, the leading O2O services platforms, and they find that the word-of-mouth effect on the consumer side and the observational learning effect on the merchant side contribute to and reinforce the expansion of a two-sided platform [3]. Dai et al. build an M/M/n queueing to discover that incentive mechanisms for both customers and crowd-sourcing drivers are crafted to improve their loyalty towards the O2O platform, in order to better manage capacity [20].

In terms of operations management in O2O business model, Hou et al. investigate the impact of investments in delivery service with one or two major competing online retailers [21]. Niu et al. (2019) find that uniform pricing always reduces the online demand size and increases the retail price, and an O2O channel leads to a high retail price and low online demand size, benefitting the retailer if the logistics cost is high [22]. In addition, some work has explored related to this issue, such as supply chain [23], omnichannel retailing [24], social interactions [25], product promotion [26], mobile payment [27] and demand delivery [28].

2.2. Personalized recommendation and similarity estimation

The personalized recommendation has become one of the main components of web technologies which help people to cope with information overload [29,30]. Based on the analysis of past consumer behavior, recommender systems filter items according to consumers' demands and interests [31,32]. These systems were first successfully used in the e-commerce business, such as Amazon [33]. Now, they have been used in many fields, including online news [34], entertainment discovery [31], energy resources [35], climate change [36], scientific exploration [37] and so on. In above research, recommendation techniques can be divided into two categories: classical methods and state-of-the-art methods. Classical method is represented by collaborative filtering (CF) [38], including service-based CF [39] and consumer-based CF [40]. Besides, many state-of-the-art computing techniques, such as graph neural network (GNN) [41], matrix factorization (MF) [42], Markov chains (MC) [43] and natural language processing (NLP) [44], have been used in the personalized recommendations and these method generally perform well.

In above recommendation methods, the preference similarity estimation is the key step which can directly determines the recommendation performance [45]. Pearson Correlation Coefficient (PCC) and Consin are common similarity measurements based on the numerical distances among consumers [31]. Considering the scale usage bias, ranking similarity becomes popular in the recommendations, including Kendall Rank Correlation Coefficient [46] and Spearman Rank Correlation Coefficient [47]. In addition, Jaccard's Coefficient has been utilized to measure the service selection similarity [4].

2.3. Summary of existing literature

In conclusion, existing research on the O2O business model predominantly emphasizes its operational aspects, with limited attention to the service selection challenges consumers encounter due to the rapid expansion of O2O services. Recommendation systems are among the most effective tools for addressing this issue. However, current methods present two key limitations: they insufficiently account for the dynamic evolution of consumer preferences in similarity analysis, and they fail to fully integrate the unique characteristics of O2O services, thereby limiting their effectiveness in O2O recommendation scenarios.

In comparison with the existing literature, the key innovations of our study are as follows: 1) Unlike static similarities used in previous research, the dynamic similarities we propose, which are based on the evolution of consumer preferences, offer a more accurate measure of relationships between consumers—especially those with strong associations—resulting in improved recommendation performance. 2) Our analysis shows that changes in selected service categories have a significant impact on the results, and using longer data time spans enhances the accuracy of measurements. 3) In contrast to traditional e-commerce,

Table 1
Changes in service category i (c_i) of oc_p and oc_q in n Time periods.

Time period								
Consumer	*1st*	*2nd*	...	j^{th}	k^{th}	...	*n-1th*	n^{th}
oc_p	$sc^1_{p,i}$	$sc^2_{p,i}$	...	$sc^j_{p,i}$	$sc^k_{p,i}$	...	$sc^{n-1}_{p,i}$	$sc^n_{p,i}$
oc_q	$sc^1_{q,i}$	$sc^2_{q,i}$	...	$sc^j_{q,i}$	$sc^k_{q,i}$	...	$sc^{n-1}_{q,i}$	$sc^n_{q,i}$
Change	$\lvert sc^1_{p,i} - sc^2_{p,i}\rvert$		...	$\lvert sc^j_{p,i} - sc^k_{p,i}\rvert$		...	$\lvert sc^{n-1}_{p,i} - sc^n_{p,i}\rvert$	

the O2O business model of 'online selection and offline use' underscores the importance of geographic service locations, which play a crucial role in consumers' service selection decisions.

3. Dynamic similarity estimation based on preference change

Preference similarity estimation is the most significant in the O2O service recommendation. Notably, the consumer's preference, including his interests and demands, may change over time due to the changes in his age, income, family, locations and many other complex factors [48]. For instance, consumers may be interested in fast food, such as KFC and McDonald's, when they are young, but may dislike them when growing up [49]. In addition, if they move house or change jobs, they may prefer to select closer O2O services to their new locations for their dietary demands. However, these changes are ignored in existing static similarity estimations. Therefore, the consumer preference change should be considered in the similarity estimation. Furthermore, consumers should have a strong relationship if they have similar preference changes over time.

In this section, we first respectively estimate dynamic similarities based on the changes in the O2O service selections, evaluations and locations over time, and then combine them together. Assume respectively $\mathbf{OC}=\{oc_i \mid i\in\{1,...,m\}\}$, $\mathbf{OS}=\{os_j \mid j\in\{1,...,n\}\}$ and $\mathbf{C}=\{c_k \mid k\in\{1,...,l\}\}$ are sets of O2O consumers, O2O services and service categories, in which n O2O services belong to k categories. $\mathbf{SR}=\{sr_{i,j} \mid i\in\{1,...,m\}, j\in\{1,...,n\}\}$ and $\mathbf{SC}=\{sc_{i,k} \mid i\in\{1,...,m\},\ k\in\{1,...,l\}\}$ respectively represent service rating matrix and service category matrix. $sr_{i,j}$ is the rating of os_j given by oc_i, and $sc_{i,k}$ is the number of services used by oc_i and belonging to category k.

3.1. Dynamic similarity estimation based on changes in service selections

O2O service selection changes can be used to analyze consumer dynamic preference [16]. O2O service platforms usually have a huge number of services [50]. For example, based on the data we collected in the Beijing area on June 1, 2024, 25,231 restaurants offer coupons to consumers on the Dianping platform. Since O2O services in the same category have high similarities, the change in one service selections cannot truly and accurately reflect consumer preference changes [51]. Therefore, we adopt changes in service categories to reflect the preference change in the estimation of dynamic similarity. Given that existing research indicates consumer preferences typically remain stable for two to three years [12], we divide the timeline into n periods, each lasting two years, during which consumer preferences are assumed to remain nearly unchanged. The following research in this paper adopts this time division.

Then, as shown in Table 1, for two consumers, oc_p and oc_q, we estimate their dynamic similarity based on the changes in the number of c_i category services they choose over n time periods, where $sc^1_{p,i}$ and $sc^1_{q,i}$ represent the respective numbers of services utilized by oc_p and oc_q belonging to category i in the *1st* time period. The evolving service selection in consecutive time periods can serve as an indicator of the dynamic evolution of consumer preferences. For oc_p, his preference change in c_i category services can be represented by $|sc^1_{p,i} - sc^2_{p,i}|$. Therefore, we can observe (n-1) such changes (i.e., $|sc^1_{p,i} - sc^2_{p,i}|$,$|sc^2_{p,i} - sc^3_{p,i}|$,..., $|sc^{n-1}_{p,i} - sc^n_{p,i}|$) in oc_p's preferences for c_i category services over n time periods, denoted as $sch^1_{p,i}$,$sch^2_{p,i}$,..., $sch^{n-1}_{p,i}$, which is analogous to oc_q's changes (i.e., $sch^1_{q,i}$,$sch^2_{q,i}$,..., $sch^{n-1}_{q,i}$). Then, we introduce Euclidean metric to estimate the dynamic similarity of consumer oc_p and oc_q in c_i category services as follows [52]:

$$ps^{c_i}_{p,q} = \sqrt{\sum_{a=1}^{n-1} sch^a_{p,i} - sch^a_{q,i}}. \tag{1}$$

Given that **SC** has l service categories, the dynamic similarity $ps^c_{p,q}$ between oc_p and oc_q based on changes in service selections can be estimated by the summary of $ps^{c_i}_{p,q}$ as follows:

$$ps^c_{p,q} = \sum_{b=1}^{l} ps^{c_b}_{p,q}. \tag{2}$$

3.2. Dynamic similarity estimation based on changes in service evaluation

Besides service selection, the service evaluation, such ratings and comments, is commonly used to analyze the consumer preference in the recommendations [53]. Evaluation similarity of two consumers is typically based on their ratings to co-used O2O services [54]. If these ratings have similar changes over time, their dynamic similarity should be very high, resulting in that one of them is likely to select services used by the other one with high evaluations. We introduce Pearson Correlation Coefficient (PCC) and Jaccard's Coefficient (JC), which are widely-used in the rating-based similarity estimation [55], in the calculation of dynamic similarity based on changes in service evaluation as follows:

$$PCC^t_{p,q} = \frac{\sum_{os_j\in \mathbf{os}^t_{p,q}} \left(sr_{j,p} - \overline{sr_p}\right)\left(sr_{j,q} - \overline{sr_q}\right)}{\sqrt{\sum_{os_j\in \mathbf{os}^t_{p,q}} \left(sr_{j,p} - \overline{sr_p}\right)^2}\sqrt{\sum_{os_j\in \mathbf{os}^t_{p,q}} \left(sr_{j,q} - \overline{sr_q}\right)^2}}, \tag{3}$$

$$JC^t_{p,q} = \frac{\left|\mathbf{OS}^{t\cap t+1}_{p,q}\right|}{\left|\mathbf{OS}^t_{p,q}\right| + \left|\mathbf{OS}^{t+1}_{p,q}\right| - \left|\mathbf{OS}^{t\cap t+1}_{p,q}\right|}, \tag{4}$$

$$ps^t_{p,q} = \begin{cases} \left|PCC^t_{p,q} - PCC^{t+1}_{p,q}\right|, & \left|\mathbf{OS}^{t\cap t+1}_{p,q}\right| = 0 \\ \left(1 + \left|PCC^t_{p,q} - PCC^{t+1}_{p,q}\right|\right)\cdot\left(1 + JC^t_{p,q}\right), & \left|\mathbf{OS}^{t\cap t+1}_{p,q}\right| \neq 0 \end{cases}, \tag{5}$$

where $PCC^t_{p,q}$ is the similarity between oc_p and oc_q in the i th time period. $\overline{sr_p}$ and $\overline{sr_q}$ are the average ratings of the services used by them, respectively. $sr_{j,p}$ is the rating of os_j given by oc_p. If he has used os_j more than once, $sr_{j,p}$ is the average rating. $\mathbf{OS}^i_{p,q}$ represents the set of services co-used by these two consumers in the t-th time period. In Eq. (4), $\mathbf{OS}^{t\cap t+1}_{p,q}$ is the intersection of $\mathbf{OS}^t_{p,q}$and $\mathbf{OS}^{t+1}_{p,q}$.

$ps^t_{p,q}$ is the dynamic similarity of oc_p and oc_q based service evaluation changes in adjacent time periods (e.g., t-th and $(t+1)$-th time periods). If co-used services of them in these two time periods are totally different (i.e., $|\mathbf{OS}^{t\cap t+1}_{p,q}| = 0$), $ps^t_{p,q}$ is the difference between PCC estimations on $\mathbf{OS}^t_{p,q}$ and $\mathbf{OS}^{t+1}_{p,q}$. Otherwise, $ps^t_{p,q}$is determined by the change in PCC estimations on the same co-used services in these two time periods (i.e., $|PCC^t_{p,q} - PCC^{t+1}_{p,q}|$) and the proportion of them to total co-used services (i.e., $JC^t_{p,q}$). As PCC falls into the interval of [0,1], $ps^t_{p,q}$is less than 1 when there are not same co-used services in adjacent time periods. In comparison, if they have same co-used services in different time periods, their preference relationship should be very closed [4]. Therefore, we introduce "enhancement factor" combination (i.e., (1+"A")(1+"B")) to ensure $ps^t_{p,q}$ is larger than 1 as the interval of JC is [0,1] as well. And the degree of $ps^t_{p,q}$ is determined by ratings of services in $\mathbf{OS}^{t\cap t+1}_{p,q}$ and the ratio of $\mathbf{OS}^{t\cap t+1}_{p,q}$ to total co-used services. Given there are n time periods, the dynamic similarity $ps^e_{p,q}$ between oc_p and oc_q based on changes in service selections can be estimated by the average dynamic similarity

calculated over n-1 adjacent time periods as follows:

$$ps^{e}_{p,q} = \frac{\sum_{t=1}^{n-1} ps^{t}_{p,q}}{n-1}, \tag{6}$$

3.3. Dynamic similarity estimation based on changes in service location

In contrast to traditional e-commerce businesses, such as Tmall and Amazon, O2O service location affects consumer decision-makings since online-purchased coupons must be used offline [56]. Therefore, consumers should evaluate usage costs on time and distance of candidate services [57]. If the used services' locations of two consumers have similar changes over time, their service demands have similarity [3]. For instance, two consumers, oc_p and oc_q, work in the same company (or live in the same area) and their daily meals mainly rely on O2O catering services. Then, if both of them leave for another company (or move to another area) about the same time, one of them prefer to select the near O2O services having been used by the other one.

Given that the consumer's personal address is unavailable, we estimate their location preference by analyzing the physical locations of the services they have utilized. Specifically, we retrieve the exact locations of these physical stores using Baidu Maps (https://map.baidu.com), represented by their latitude and longitude coordinates. Subsequently, we determine the geographic center of these stores by calculating the mean of their latitude and longitude coordinates, which serves as the consumer's location preference [4]. For two consumers, denoted as oc_p and oc_q, once their location preferences are established, we employ the Haversine formula to calculate the distance between them, represented as $d_{p,q}$ (km). The Haversine formula is a widely adopted method for computing the distance between two points based on their latitude and longitude coordinates [58].

Given there are n time periods, we can get n values of $d_{p,q}$. Therefore, the dynamic similarity $ps^{l}_{p,q}$ between oc_p and oc_q, based on changes in service locations, can be estimated by the average distance calculated over n time periods as follows:

$$ps^{l}_{p,q} = \frac{\sum_{t=1}^{n} d^{t}_{p,q}}{n}. \tag{7}$$

4. O2O service recommendation based on dynamic similarity

In this section, we first integrate the three dynamic similarities discussed in Section 3. Then, for a target consumer, we employ two distinct similarity-based recommendation methods using the proposed dynamic similarity for service recommendations.

4.1. The combination of dynamic similarities

The data generated by consumers interacting with O2O services is critical for analyzing consumer preferences. The types of services selected by consumers can directly reflect their explicit preferences at a macro level [59]. In contrast, consumer reviews provide insights into their latent preferences at a micro level [60]. Additionally, unlike traditional e-commerce services, O2O service choices are distance-sensitive, meaning the locations of the services used can indicate consumers' location preferences [57]. In fact, consumer similarities can be independently analyzed based on their preferences on service selections, service evaluations, and the geographic locations of chosen services. Existing literature on recommendation systems has also utilized each of these preferences individually to analyze consumer similarities [60–62]. Although these preferences reflect consumer demand from different perspectives, the similarities based on them can complement each other, and crucially, they exist within the same dimension [63]. For such different consumer similarities within the same dimension, existing literature employs a linear function with weight parameters to combine them [4,6,64]. Following this approach, we also use this method to combine the three dynamic similarities based on changes in these features over time as follows:

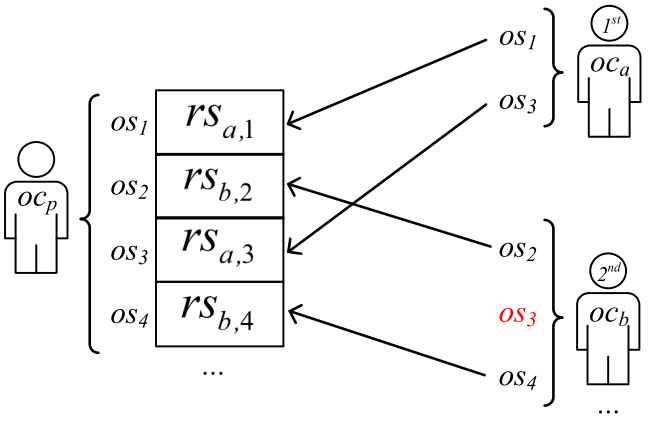


Fig. 1. An example of predicted rating generation.

$$ps_{p,q} = \alpha * ps^{s}_{p,q} + \beta * ps^{e}_{p,q} + (1-\alpha-\beta) * ps^{l}_{p,q}, \tag{8}$$

where α and β are weight parameters for adjusting influences that $ps^{s}_{p,q}$, $ps^{e}_{p,q}$ and $ps^{l}_{p,q}$ have in this combination, whose interval is [0,1]. In the experiments described in Section 5.3.2, we adjust the values of α and β to identify their optimal settings by observing the recommendation performance under different values, and further interpret the managerial significance of these optimal values.

4.2. O2O service recommendation

In the preceding section, we compute the dynamic similarities among consumers. In this section, we explore two distinct similarity-based recommendation methods: the classical collaborative filtering recommendation method (CF-*Re*) and the advanced neighbor-dominated recommendation method (ND-*Re*). We substitute the traditional similarity metrics in these two methods with our proposed dynamic similarity method, leading to the development of two dynamic similarity-based recommendation methods (ReDPS): ReDPS-CF and ReDPS-ND.

4.2.1. ReDPS-CF: collaborative filtering recommendation method based on dynamic similarity

In similarity-based recommendation systems, collaborative filtering recommendation (CF-*Re*), which utilizes consumer behavior and preferences to make personalized recommendations by identifying similarities among consumers or items, is considered the most classic approach and is widely adopted in business practices [65]. This method involves three steps: similarity estimation, neighbor selection, and rating prediction [66]. The predicted rating for a candidate service os_i of a target consumer oc_p can be calculated in the following:

$$sr^{pre}_{p,i} = \overline{sr_p} + \frac{\sum_{q \in N_{p,i}} ps_{p,q} \times \left(sr_{q,i} - \overline{sr_q}\right)}{\sum_{q \in N_{p,i}} ps_{p,q}}, \tag{9}$$

where $\mathbf{N_{p,i}}$ represents the set of oc_p's neighbors who have utilized os_i, and $\overline{sr_p}$ and $\overline{sr_q}$ are the average ratings of O2O services consumed by oc_p and oc_q, respectively. Once the ratings for all candidate services of oc_p are estimated, they are ranked in descending order based on the predicted ratings. The top-ranked services are then recommended to oc_p in line with the platform's business requirements.



Fig. 2. An example in Dianping dataset.

4.2.2. *ReDPS-ND: neighbor-dominated recommendation method based on dynamic similarity*

In addition, a relatively advanced similarity-based recommendation method compared to CF, ND-*Re*, can be used in this research for recommendation. This method bases its suggestions on evaluations of candidate services by his neighbors and their respective rankings, and its core principle is that higher-ranked neighbors should exert more influence on service recommendations for the target consumer [4,67].

Fig. 1 illustrates an example of ReDPS-ND. For a target consumer oc_p, os_1, os_2, os_3 and os_4 are his candidate O2O services, and oc_a and oc_b are his neighbors, ranked first and second based on their dynamic similarities with him. For a candidate service (e.g., os_1), its predicted rating for oc_p is determined by the rating (i.e., $sr_{a,1}$) given to this service by oc_p's highest-ranked neighbor (i.e., oc_a) who has used this service. For instance, even if both oc_a and oc_b have used os_3, as shown in Fig. 1, the predicted rating for this candidate service is $sr_{a,3}$ rather than $sr_{b,3}$, because oc_a ranks higher than oc_b. Then, candidate services are sorted in descending order based on predicted ratings, and the O2O service platform can recommend several top of them to the target consumer according to its business requirements.

5. Experiments

In this section, we first introduce the two datasets used in the experiments: our captured Dianping dataset and the Yelp open dataset. Second, we describe three performance metrics utilized in the experiments, as well as the cross-validation method employed to reduce uncertainty and enhance the reliability of the experimental results. Third, we present and analyze the experiment results.

5.1. Datasets

To enhance the reliability and stability of our experiments, we conduct experiments on two O2O service datasets from different countries, obtained through distinct channels. Specifically, we use the Dianping dataset, which we collected from the Dianping platform in China, and the Yelp open dataset, provided by the Yelp platform in the United States.

5.1.1. *Dianping dataset*

Dianping is one of famous and widely-used O2O service platforms in China. On this platform, catering group buying is the most popular O2O service. A large number of Chinese people are used to viewing coupons

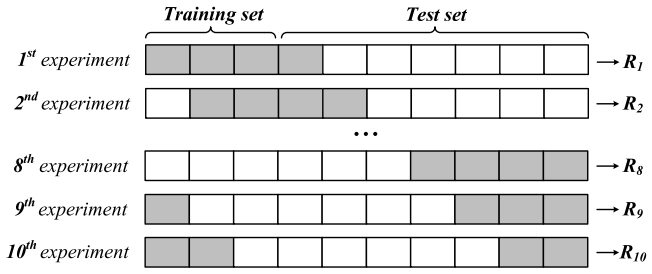


Fig. 3. An example of improved 10-*cv*.

provided by restaurants on Dianping when making decisions in their service selections [64]. Due to the strong regional nature of O2O services, we collected group-buying data for catering establishments in Beijing, the capital of China with over 25,000 restaurants, from the Dianping platform. The data spans 14 years, from January 3, 2011, to June 1, 2024, and was collected in June 2024.

As shown in Fig. 2, we use a real data example from the Dianping dataset, captured from the Dianping official website (https://www.dianping.com), to illustrate the dynamic feature of service evaluation in this dataset. In this example, consumer "dpuser_2,654,681,050″ used and reviewed the same dining service, "Jubaoyuan (Chaoyang)", on November 30, 2021, and again on November 11, 2023. Over these two years, the review changed from an initial "five-star" to a "half-star", reflecting his preferences change. Furthermore, Fig. 2 also illustrates the essential data fields required for the experiments in this research, encompassing service information (i.e., service name, service category, and service location) and review information (i.e., consumer name, review time, and service rating). We capture 1004,891,109 reviews given by 882,310 consumers to 25,830 O2O services belonging to 43 service categories. We transform these data into an 882,310×25,830 rating matrix and an 882,310×43 category matrix with the utilization of Java (Version: 21.0.3) and MATLAB (Version: R2024a). The values in the rating matrix range from 0 to 5 with intervals of 0.5, where a rating of 0 indicates no rating given. Values in the category matrix are integers from 0 to 47.

5.1.2. Yelp open dataset

Yelp platform, founded in 2004 in America, is an O2O platform recognized for its consumer-generated reviews and ratings, similar to Dianping, with a particular focus on catering services. It facilitates informed consumer decisions by providing community feedback on restaurants and other local businesses. Yelp platform provides an open dataset (https://www.yelp.com/dataset) that includes consumer profiles, service information, and review content. In detail, this dataset encompasses 6990,280 reviews provided by 1987,929 consumers for 150,346 services over an 18-year period from February 16, 2005, to January 19, 2022. The dataset includes comprehensive information crucial for our experimental study, such as service name, service category, service location, consumer name, review time, and service rating (https://www.yelp.com/dataset/documentation/main).

5.2. Recommendation performance metrics and cross-validation

We utilize precision, recall, and F1-score as evaluation metrics for recommendation performance, which are commonly employed in research on recommender systems [68].

In the experiments, we utilize precision, recall, and F1-score as evaluation metrics for recommendation performance, which are commonly employed in research on recommender systems [68]. In addition, 10-fold cross-validation (10-*cv*) is used to check the robustness of results and reduce uncertainty, thereby enhancing their reliability [69,70]. Due to our experiments conducted with varying data volumes, we utilize an improved 10-*cv* method [50], as illustrated in an example in Fig. 3. Initially, the dataset is randomly divided into 10 mutually exclusive subsets (i.e., 10 blocks) of approximately equal size. Instead of selecting just one subset as usual, we sequentially chose n subsets, where n is greater than 0 and less than 10, for training, while the remaining subsets are used for testing the method. In this example, n equals 4 (i.e., 4 gray blocks), indicating that 40 % of the experimental data is used for the training set, while the remaining 60 % (i.e., 6 white blocks) is allocated to the test set. As shown in Fig. 3, we conduct ten experiments (i.e., "10-fold"), resulting in ten experimental outcomes R_i ($i\in[1,10]$), ensuring that each subset is used at least once as both training and testing data. Finally, to check robustness, we calculate the standard deviation of the 10 experimental results as follows [71]:

$$s_R = \sqrt{\frac{\sum_{i=1}^{10}(R_i - \overline{R})}{9}} \tag{10}$$

where R_i represents the result of the i th experiment, and $\overline{R}$ denotes the mean of the 10 experimental results. If S_R falls below the set robustness threshold, we utilize $\overline{R}$ as the final experimental result to enhance the reliability of the findings.

5.3. Experiments

Based on above research, we propose three research questions to guide our experiments and discussions: 1) Can dynamic similarity, measured based on temporal changes in consumer preferences, more accurately reflect relationships among consumers? 2) To what extent do changes in consumers' service selection, service evaluation, and the location of chosen services influence the formation of O2O consumer preferences? 3) Do static consumer attributes, such as gender, impact O2O service recommendations, and if so, to what extent?

Therefore, in experiments, we first compare proposed ReDPS methods, ReDPS-CF and ReDPS-ND, with classical and state-of-the-art recommendation methods. As collaborative filtering (CF) has been successfully used in recommender systems for more than 20 years [72], three classical CF-based methods, item-based CF (I-CF) [73], user-based CF (U-CF) [74] and hybrid-based CF (H—CF) [72], are selected in comparison experiments. In addition, with the fast development of computer science, many state-of-the-art technologies have performed well in recommendations [75], such as recurrent neural network (RNN) [76] matrix factorization (MF) [42], Markov chains (MC) [43] and natural language processing (NLP) [44]. Given the respective time spans of 14 years for the Dianping data and 18 years for the Yelp data, experiments with ReDPS are conducted on these datasets with the time period parameter n set to 7 and 9, respectively.

Second, we conduct an in-depth study of the proposed ReDPS, including: 1) Comparing ReDPS-CF and ReDPS-ND across various time periods (i.e. parameter n), assessing their recommendation effectiveness and method efficiency, and analyzing the influence of n on their performance; 2) testing various values of α and β in the ReDPS methods to determine the optimal values that yield the best algorithm performance and analyzing the significance of these optimal values; 3) further exploring the impact of additional consumer characteristics (e.g., age, gender) on the analytical results.

The experiments are conducted on a server equipped with an Intel Xeon Gold 6530 CPU running at 2.10 GHz with 32 cores, 512 GB DDR4 RAM, and an NVIDIA Tesla A100 GPU with 40G VRAM. The operating system used is Ubuntu 20.04 LTS.

5.3.1. Comparison experiments

In our experiments, we first conduct external comparative experiments to assess the recommendation effectiveness of the two proposed ReDPS methods in conjunction with seven comparative methods across the Dianping and Yelp datasets, varying the data volumes. Subsequently, we perform internal comparative experiments focusing on the ReDPS

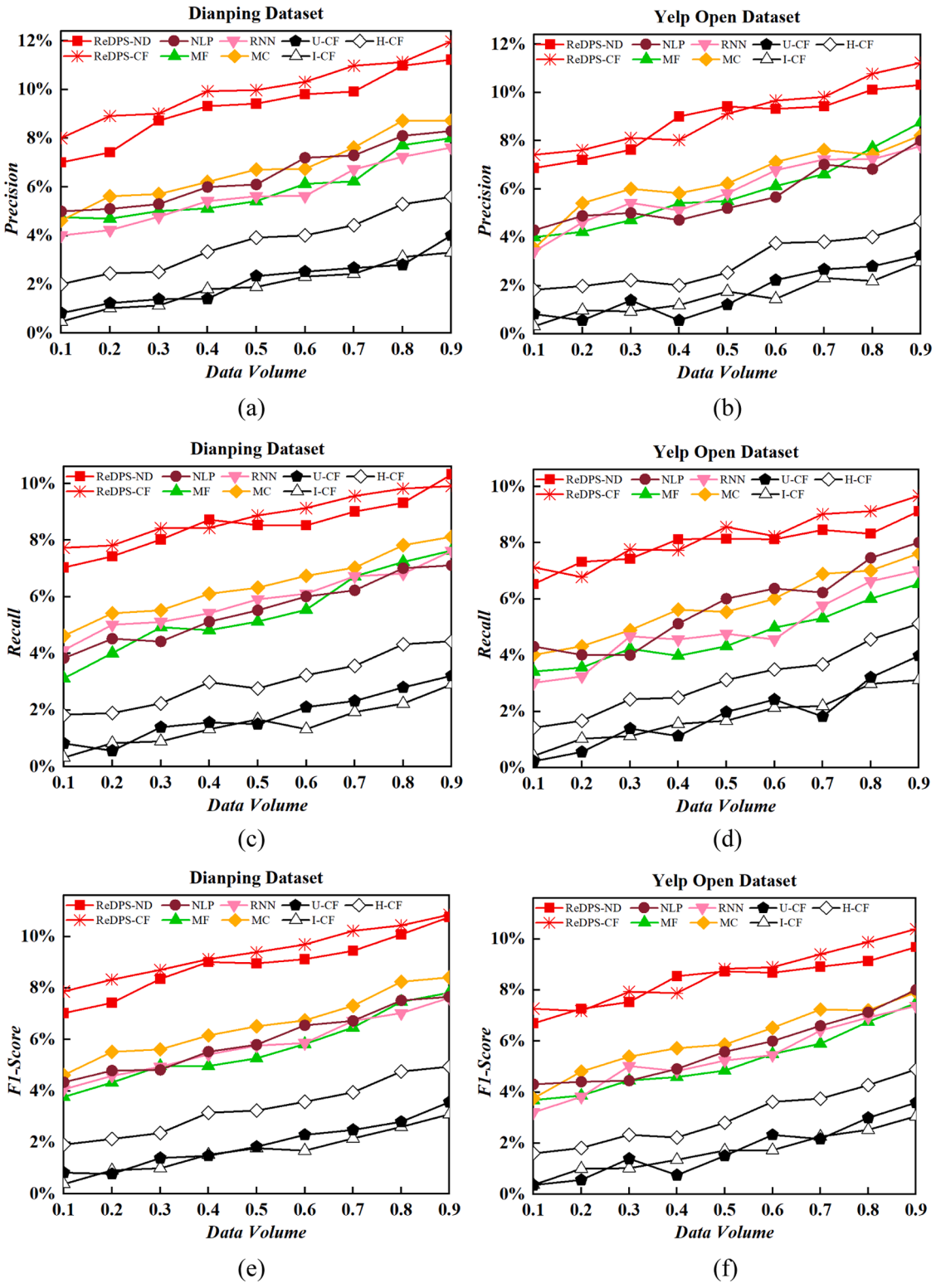


Fig. 4. The recommendation method performances in different data volume.

methods, evaluating the recommendation performance of ReDPS-CF and ReDPS-ND under different time period parameters n.

(**1**) External comparative experimental results

In the experiments, ratings ranging from 10 % to 90 % (in increments of 10 %) are randomly extracted as training sets, with the remainder used as testing sets. Additionally, Top-k is set to 30, indicating the number of recommended services per customer, which is a common default setting in most recommendation research [68,77]. Both α and β are uniformly set to 0.33 in this section, with a detailed analysis provided in Section 5.3.2.

Fig. 4 illustrate the performances of recommendation methods evaluated by recommendation precision, recall and F1-Score on the

Table 2
The performance of ReDPS-CF and ReDPS-ND with different time periods n on the Dianping dataset.

Time Periods								
Metrics		**1**	**2**	**3**	**4**	**5**	**6**	**7**
F1-Score	**ReDPS-CF**	2.85 %	6.92 %	8.25 %	9.03 %	9.56 %	9.82 %	10.43 %
	ReDPS-ND	1.62 %	6.78 %	7.98 %	8.32 %	8.49 %	9.37 %	10.12 %
Running Time	**ReDPS-CF**	0.192 s	0.283 s	0.316 s	0.442 s	0.772 s	1.309 s	1.550 s
	ReDPS-ND	0.136 s	0.167 s	0.198 s	0.202 s	0.324 s	0.433 s	0.698 s

Table 3
The performance of ReDPS-CF and ReDPS-ND with different time periods n on the Yelp open dataset.

Time Periods									
Metrics		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
F1-Score	**ReDPS-CF**	1.56 %	6.42 %	7.29 %	7.89 %	8.22 %	8.91 %	9.00 %	9.42 %	10.33 %
	ReDPS-ND	1.03 %	6.09 %	7.01 %	7.35 %	7.98 %	8.07 %	8.32 %	9.38	9.61 %
Running Time	**ReDPS-CF**	0.039 s	0.047 s	0.098 s	0.124 s	0.188 s	0.452 s	0.793 s	1.234 s	1.311 s
	ReDPS-ND	0.012 s	0.024 s	0.055 s	0.089 s	0.103 s	0.235 s	0.300 s	0.561 s	0.840 s

Dianping and Yelp datasets, respectively. Since, in general, more data contains more information, which can lead to more accurate predictions, this performance verifies the validity and reliability of the Dianping and Yelp datasets [65,77]. Furthermore, it demonstrates that proposed ReDPS methods and the comparative methods are effective for recommendations on these two datasets [4,66].

In detail, the proposed ReDPS methods significantly outperform other recommendation methods. These results indicate that consumer preferences change over time. Furthermore, if consumers exhibit similar changes, they have strong similarity relationships. Consequently, one consumer is likely to select services previously used by others with similar preferences. Therefore, dynamic similarity based on preference change is a strong measure of similarity and can accurately estimate consumer relationships. In comparison, the other seven tested recommendation methods do not consider the influence of time. A detailed discussion of ReDPS-ND and ReDPS-CF will be provided in Section 5.3.1 (2).

In addition, recommendation methods based on state-of-the-art computer technologies, RNN, MF, MC and NLP, have similar performance to one another, and are superior to CF-based methods, I-CF, U-CF and H—CF. Since data sparse is a common problem in recommendations [6], CF-based methods perform poorly due to the inaccurate similarity estimation based on the ratings of co-used services of two consumers [78]. In comparison, state-of-the-art methods can deliver better performances by repeated training to accurately obtain the customer preference [76]. However, their spatiotemporal complexities are much higher than those of CF-based methods. Thus, most internet companies prefer to more stable and lower cost CF-based methods rather than state-of-the-art recommendation methods [79].

(2) Internal comparative experimental results

We further analyze the recommendation effectiveness and efficiency of the proposed ReDPS methods, ReDPS-CF and ReDPS-ND. Specifically, we observe the performance of these two methods, evaluated by F1-Score (i.e., effectiveness) and running time (i.e., efficiency), on the Dianping and Yelp datasets under the same computational environment while adjusting the time period parameter n. Given that the Dianping dataset spans 14 years (from January 3, 2011, to June 1, 2024) and the Yelp open dataset spans 18 years (from February 16, 2005, to January 19, 2022), the maximum allowable values for n are 7 (i.e., “14/2″) for Dianping and 9 (i.e., “18/2″) for Yelp when using 2-year intervals.

The experimental results in Tables 2 and 3 indicate that both ReDPS methods demonstrate an increasing F1-Score as n increases across both datasets. However, this improvement is accompanied by longer running times for these two methods. Notably, when $n = 1$, both methods exhibit significantly lower effectiveness compared to their performance at other n values. Additionally, at equivalent n values, the methods exhibit longer running times on the Dianping dataset compared to the Yelp open dataset, primarily due to the larger data volume in the Dianping dataset. The experimental results mean that as the dynamic evolution of consumer preferences over time spans can be better captured, it becomes possible to more accurately assess the similarity relationships between consumers. Furthermore, the poor performance at $n = 1$ (i.e., the static similarity) suggests that traditional similarity measures, in contrast to dynamic ones, inadequately capture the association between consumers. Additionally, increasing n results in higher algorithmic complexity, thereby reducing computational efficiency. Under the same conditions, ReDPS-CF demonstrates slightly better recommendation effectiveness compared to ReDPS-NG. However, ReDPS-CF has significantly lower computational efficiency due to the higher complexity of the collaborative filtering framework compared to the neighbor-dominated approach. Therefore, when sufficient computational resources are available, ReDPS-CF is preferred; otherwise, ReDPS-NG is a more

Table 4
Standard deviation of 10-*cv* in F1-Score generation for recommendation methods on Dianping/Yelp datasets (1E-5).

Method Data volume	**ReDPS-ND**	**ReDPS-CF**	**NLP**	**MF**	**RNN**	**MC**	**U-CF**	**I-CF**	**H-CF**
0.1	3.2/4.8	9.2/4.0	3.9/5.5	3.2/1.9	0.4/8.2	9.1/3.0	1.0/2.4	0.8/4.2	1.9/2.3
0.2	5.0/4.2	1.9/8.0	4.4/5.9	3.2/9.0	4.2/6.7	1.3/8.7	1.1/9.0	7.8/3.2	9.4/4.4
0.3	3.6/1.2	9.0/5.2	3.9/3.0	8.9/3.3	5.4/6.9	1.0/9.8	0.6/4.5	8.9/9.2	2.8/1.7
0.4	6.8/3.2	0.3/5.1	0.2/3.3	9.8/2.0	1.8/4.5	7.2/3.9	5.5/6.8	9.3/7.7	2.0/2.9
0.5	1.8/9.7	4.8/9.0	4.3/8.5	3.2/8.8	0.4/5.7	1.9/1.0	6.1/3.4	5.0/4.4	3.6/9/1
0.6	2.3/4.4	3.8/8.3	9.2/1.0	2.2/4.5	9.2/5.0	6.6/3.9	4.1/7.8	1.0/0.8	3.3/6.7
0.7	5.5/9.0	8.8/5.3	1.9/0.5	2.2/3.9	4.4/6.1	1.9/2.9	3.4/9.3	0.6/2.6	5.5/9.5
0.8	8.2/4.4	4.0/8.8	4.5/9.4	4.3/8.4	1.9/4.4	7.8/9.5	7.6/7.7	5.6/3.9	2.2/1.0
0.9	0.9/4.2	3.2/9.0	8.9/3.1	5.2/4.4	7.8/2.0	1.0/0.5	5.5/9.2	8.7/8.8	9.3/3.0

Table 5
The significance of parameters in ReDPS methods.

Parameter	Significance
α	Similarity estimation based on service **category** change
β	Similarity estimation based on service **rating** change
$(1-\alpha-\beta)$	Similarity estimation based on service **location** change

suitable choice.

(3) Robustness **test**

As F1-Score is the combination of recommendation precision and recall, we present the robustness tests of F1-score values (i.e., Fig. 4(e) and (f)) in Table 4. Each set of two numbers in the table represents the standard deviation of ten experimental results generated by 10-*cv* on the Dianping and Yelp datasets, respectively. For example, "3.2/4.8″ in the first row and first column indicates that, for ReDPS-ND under the condition of data volume 0.1, the standard deviation of the ten experimental results generated by 10-*cv* is 3.2 for the Dianping dataset and 4.8 for the Yelp dataset.

The experimental results depicted in Fig. 4(e) and (f) report F1-Score values ranging from 0 to 0.12 (i.e., 0 % to 12 %), presented with a precision of four decimal places (1E-4). This meticulous precision effectively discerns the recommendation performance of the methods under evaluation. Consequently, we established 1E-4 (i.e., 0.0001) as the threshold for robustness testing. If the standard deviation of the ten experimental results obtained from 10-*cv* falls below this threshold, the robustness test is passed, and the mean of these ten results is used as the final experimental result.

Table 4 demonstrates that all standard deviations are within the magnitude of 1E-5, indicating that they are all less than 1E-4. Therefore, Therefore, the results of the 10-*cv* experiments have successfully passed the robustness test, and the means of these results constitute each data point in Fig. 4(e) and (f). Subsequent experimental results have also passed the robustness test. However, due to the space limitation, the detailed robustness test results cannot be fully presented in the paper.

5.3.2. *Parameter experiments*

The proposed ReDPS methods introduce two parameters, α and β, to determine the influence of three dynamic similarities respectively based on service selection, evaluation, and location in the combination. As shown in Table 5, α and β represent the weights of dynamic similarities reflecting changes in consumer preferences related to service categories and ratings over time, respectively, while $(1-\alpha-\beta)$ represents the weight of the dynamic similarity based on changes in service location preferences. We select ReDPS-CF, which demonstrates better recommendation effectiveness than ReDPS-NG, for parameter testing on Dianping and Yelp datasets with Top-N setting of 30. In the experiments, we simultaneously adjust these two parameters from 0 to 1 in increments of 0.001. It is essential to maintain throughout the experiment that the sum of α and β does not exceed 1, given that the weights of all three similarities are equal to or greater than 0.

Fig. 5 illustrates recommendation performance, evaluated by F1-Score, of ReDPS-CF with various combinations of α and β on Dianping and Yelp datasets, respectively. In details, both Fig. 5(a) and (b) exhibit a "mountain" shape, with dashed lines connecting the peak points, representing optimal recommendation performance, to their respective parameter values. ReDPS-CF achieves optimal recommendation performance at $\alpha = 0.542$ and $\beta = 0.109$ for the Dianping dataset, and at $\alpha = 0.421$ and $\beta = 0.153$ for the Yelp dataset. Notably, the two mountains' edges, which indicate that one or both weights are equal to 0, are very steep, meaning the corresponding F1-Scores are significantly lower than those generated by other parameter combinations. The lowest points of the mountains, where $\alpha = 0$ and $\beta = 1$, means that ReDPS-CF has the worst recommendation performance.

These results indicate that service selection (i.e., α), measured by service category, holds greater importance in measuring consumer preferences. In contrast, service evaluation (i.e., β), reflected in ratings, is more detailed and susceptible to various influencing factors such as consumer mood, weather conditions, and traffic [80], thereby not accurately reflecting changes in consumer preferences. Additionally, since consumer choices of O2O services are sensitive to location [3,57], changes in the locations (i.e., $(1-\alpha-\beta)$) of used O2O services signify shifts in consumer demand preferences. Nevertheless, dynamic similarity among consumers is determined by all changes in service selection, evaluation, and location, and neglecting any of these factors significantly affects the estimation accuracy.

5.3.3. *Further discussion of impact of consumer gender in ReDPS methods*

The previous experimental results demonstrate the effectiveness of the proposed dynamic similarity in accurately capturing consumer relationships, leading to the satisfactory recommendation performance. Beyond dynamic similarities based on changes in service selection, evaluation, and location, static consumer attributes such as gender significantly influence recommendation outcomes [81]. In the

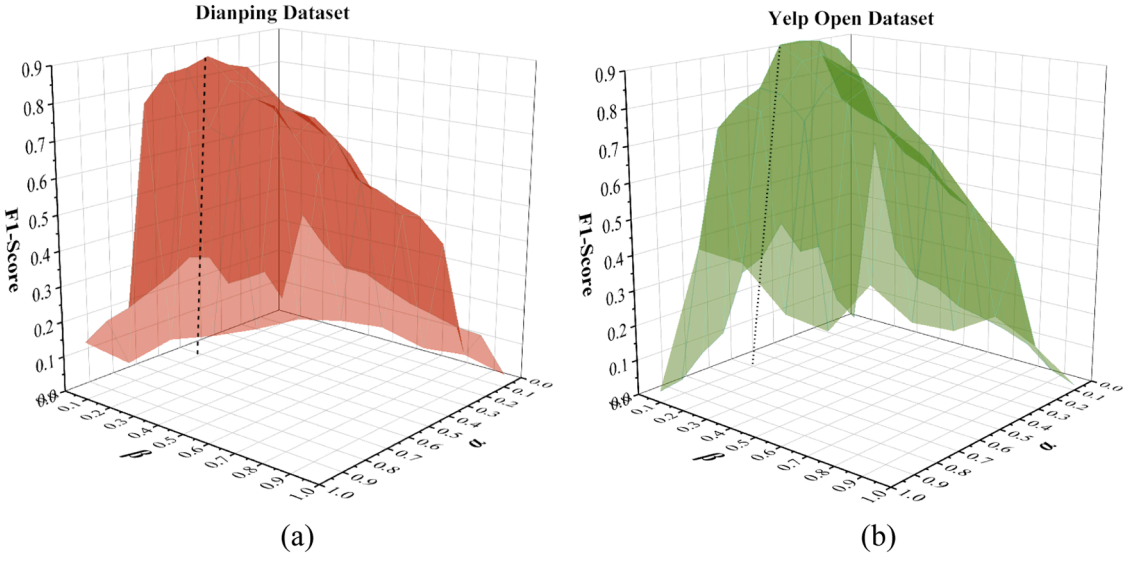


Fig. 5. The performances of ReDPS-CF with different combinations of α and β.

Table 6
The performance of ReDPS methods with different gender strategies in recommendation (F1-Score, %).

Data volume										
Method		**10 %**	**20 %**	**30 %**	**40 %**	**50 %**	**60 %**	**70 %**	**80 %**	**90 %**
ReDPS-NG	**Origin**	7.02	7.43	8.36	9.01	8.95	9.12	9.44	10.08	10.76
	Sameness	6.85	7.23	8.87	9.33	9.45	10.00	10.31	10.98	11.24
	Difference	2.38	2.89	3.03	3.22	3.47	3.68	4.02	5.66	5.82
ReDPS-CF	**Origin**	7.87	8.33	8.71	9.12	9.39	9.69	10.22	10.44	10.85
	Sameness	8.11	8.92	9.01	9.34	9.87	10.38	11.24	11.15	12.09
	Difference	1.62	1.82	2.31	3.01	3.67	4.00	4.72	4.36	5.03

experiments, we explore how consumer gender impacts the performance of our proposed ReDPS methods, ReDPS-CF and ReDPS-NG. Since the accuracy of neighbor selection directly influences the final recommendations for the target consumer, we integrate gender considerations into this process. As shown in Table 6, we propose three gender strategies in the recommendation: "Origin", "Sameness", and "Difference". "Origin" disregards gender influence in neighbor selection. "Sameness" selects neighbors of the same gender as the target consumer (i.e., males recommend to males, females recommend to females), while "Difference" selects neighbors of the opposite gender (i.e., males recommend to females, females recommend to males).

As shown in Table 6, we highlight the superior recommendation performance of ReDPS-NG and ReDPS-CF under identical data volumes when employing different gender strategies. The experimental results unexpectedly reveal that for both ReDPS-CF and ReDPS-NG, the "Sameness" strategy performs slightly better than the "Origin" strategy, which disregards gender, and significantly outperforms the "Difference" strategy. These findings underscore the significant role of gender in O2O service recommendations, indicating that consumers of the same gender share similar preferences in specific aspects [82]. Conversely, substantial differences in O2O service preferences exist between consumers of different genders [83].

6. Conclusion

In this paper, we propose a novel O2O service recommendation method, ReDPS, based on dynamic preference similarity. Consumer preference changes are reflected in alterations to the service selection, evaluation, and location of the services they use. Service selection changes are measured by service categories rather than individual services. Consumer ratings for the same services across different time periods are used to estimate changes in service evaluation. Location changes are determined by the distance between the central locations of the consumer's used services over time. Using these preference changes, we estimate dynamic similarity to assess consumer relationships. Subsequently, for a target consumer, we employ this similarity to rank neighbors and recommend O2O services.

The experimental results demonstrate that the proposed dynamic similarity accurately reflects strong consumer relationships. Specifically, the dynamic similarity analysis extends the traditional focus on static consumer preferences at a single stage to incorporate dynamic preferences across multiple stages. This expansion enhances our understanding of robust consumer relationships, thereby enriching theories related to consumer preference and relationship analysis in recommender systems. Practically, these findings offer O2O service platforms several advantages: implementing the ReDPS method enables platforms to achieve superior recommendation performance compared to traditional approaches, fostering greater consumer trust and increasing revenue. Furthermore, platforms can tailor methods and time spans based on their computational capabilities, with ReDPS-CF outperforming ReDPS-NG, particularly over extended periods when sufficient resources are available. Lastly, prioritizing consumer gender in service recommendations can improve accuracy, as consumers of the same gender often exhibit similar preferences.

Even though this research presents significant novelties, it also has several limitations that we highlight here: 1) The proposed method does not incorporate textual, image, or video information from comments in the consumer preference analysis. 2) Besides gender, other essential consumer attributes, such as age, are not publicly available and thus cannot be obtained through data scraping. Additionally, the Yelp Open dataset used in this study does not include these attributes. 3) Since we cannot access real-time geographic location data of consumers, the real-time location factor, which significantly influences O2O service recommendation effectiveness, cannot be considered in the proposed method. Therefore, in future studies, we should first incorporate multi-source heterogeneous data, such as text and images from comments, into the preference similarity estimation to enhance method performance. Second, we aim to collaborate with leading O2O platforms to obtain more non-public and real-time data, thereby expanding the applicability of our method.

CRediT authorship contribution statement

Lu Xu: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Conceptualization. **Yuchen Pan:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Desheng Wu:** Supervision, Conceptualization. **David L. Olson:** Writing – original draft, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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